

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)	I	8.2 kΩ (+/- 0.2: reading from the candidate's graph) (1) 8.2 = 12 / current (substitution) (1) Current = 12 / 8.2 (Manipulation) (1) = 1.46 / 1.5 mA (1) Manipulation mark can be awarded for current = voltage / resistance $I = V/R$ If 8.4 kΩ is read off graph: Expect answer of 1.43 / 1.4 mA (4) If 8.6 kΩ is read off graph: Expect answer of 1.4 / 1.395 mA (4)		4		4	4	4
			II	0.00146 A (ecf) (1) conversion of candidate answer in part (I) above $\times 12 = 0.0175 / 0.0174W$ (1) Or $(0.00146)^2$ (ecf) $\times 8.2 \times 10^{-3}(1) = 0.0174/0.0175W$ (1) Accept 17.52 for (1)		2		2	2	2
	(c)			Test at lower temperatures (1) obtaining lower temperature using ice (1)			2	2		2
				Question 4 total	0	14	2	16	9	16